

☰ PHASE 1 · 2012–2020

## Disembodied ML

The Cloud Era

**Systems Focus:**

High-throughput digital predictions ( $x \rightarrow y$ )

- **Substrate:** Cloud GPU Clusters
- **Workloads:** ResNet, BERT, RecSys
- **Boundary:** Stateless API / Screen
- **Failure Mode:** `try/catch` → Retry
- **Physical Action:** None (Bits)

✂ Compress

🖨 PHASE 2 · 2018–2023

## Edge Perception

The TinyML Era

**Systems Focus:**

Compressing models onto MCUs

- **Substrate:** Bare-Metal MCUs / DSPs
- **Workloads:** Wake-words, Anomaly
- **Boundary:** Open-loop sensing
- **Failure Mode:** Dropped alert
- **Physical Action:** Passive telemetry

🧠 Reason

🧠 PHASE 3 · 2023–2026

## Deliberation

The Foundation Era

**Systems Focus:**

Spatial reasoning using foundation models

- **Substrate:** Edge MPUs / NPUs
- **Workloads:** VLMs, Transformers
- **Boundary:** Semantic planning (1 Hz)
- **Failure Mode:** Hallucination,  $P_{99}$  tails
- **Physical Action:** Untrusted proposals

🔄 Close Loop

🏠 PHASE 4 · NOW (FRONTIER)

## Physical AI Systems

Closed-Loop Actuation

**Systems Focus:**

Proposals governed by real-time enforcers

- **Substrate:** Linux MPU + Real-Time MCU
- **Workloads:** Multi-Rate VLA + 1 kHz CBF
- **Boundary:** Delegated authority
- **Failure Mode:** Gearbox shear / Impact
- **Physical Action:** Matter & kinetic energy

← 🖥 Open-Loop & Digital Sandboxes (Software retries, idempotent computation, no motor coils)

⚙ Physical Causality ( $W_t \rightarrow W_{t+1}$ , No ctrl+z) →